Covariance and Correlation

→ Covariance is a statistical term that refers to a systematic relationship between two random variab

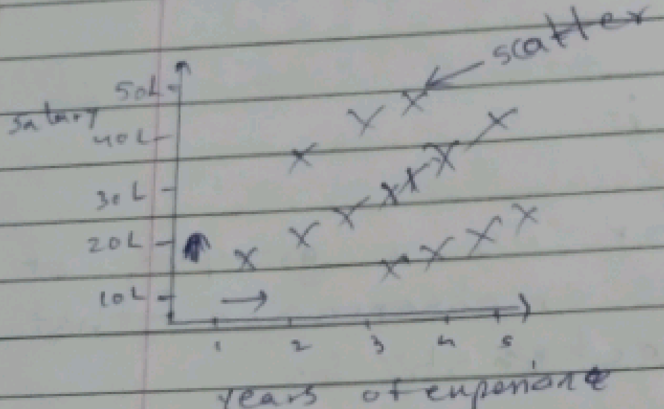
→ both variable/value Increase

→ In which a change in the other reflect a change in one variable.

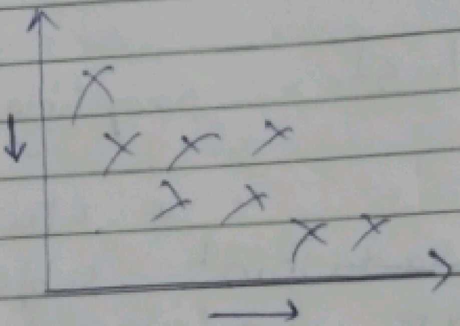
The range of covariance is between $-\infty$ to $+\infty$

$+30, +300, +3000, \ldots\ldots +\infty$

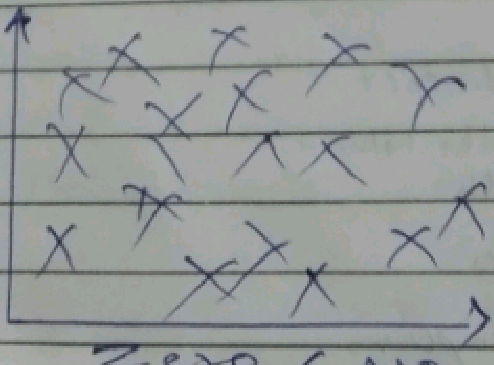
$-30, -300, -3000, \ldots\ldots -\infty$



+ covariance



- ve Covariance



Zero / No Variance

Disadvantage of Range $-\infty$ to $+\infty$

Correlation

If one variable increase dose the other variable increase or decre--ase

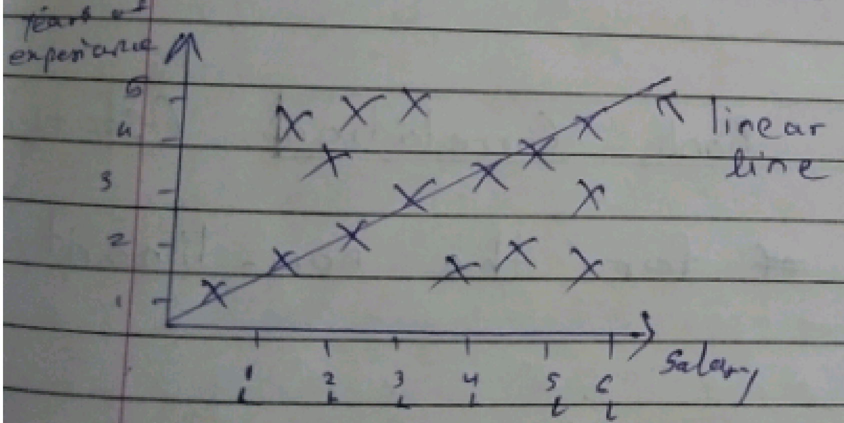
↑↓ / ↑↑

→ The range of correlation is between −1 to +1
−100% +100%.

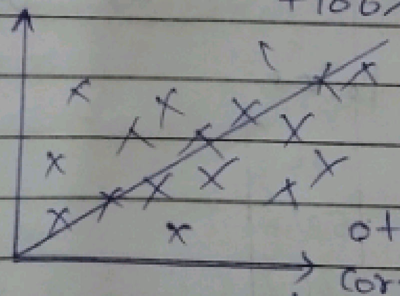
① Pearson correlation

→ It is the most common way of measuring a linear correlation.

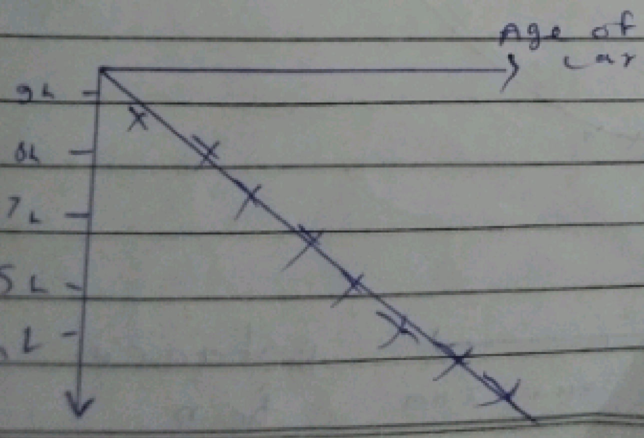
(linier line
$y = mx + c$



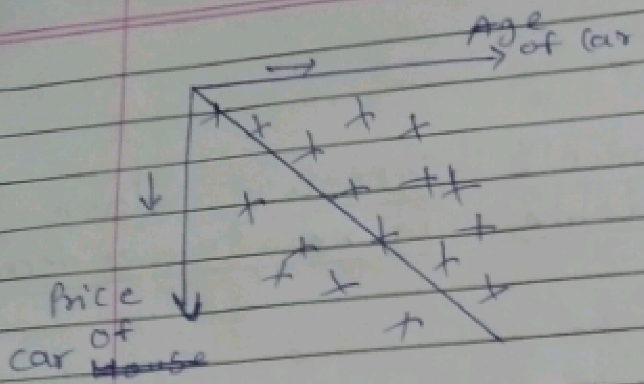
Salary ↑ year ↑
+ve correlation
+100%



0 to 1
correlation
moderate
positive

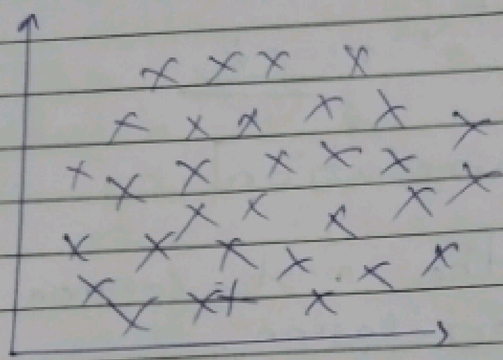


Strong Negative
−100% (−1)



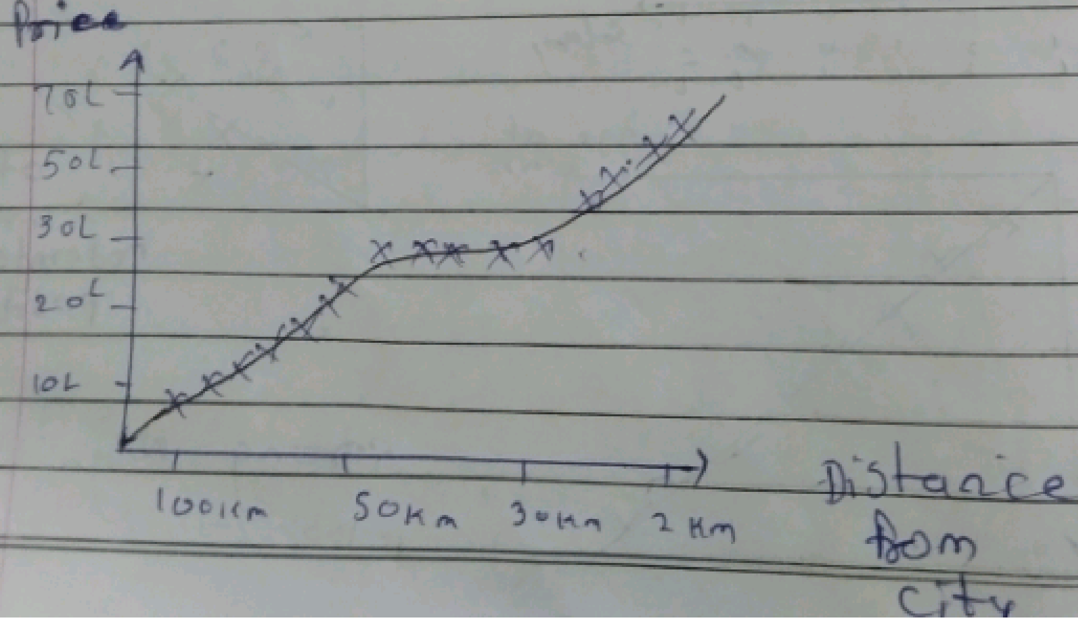
moderate negative
(-1 to 0)

zero correlation
(0)



2) $\boxed{\text{Spearsman's Rank Correlation}}$ (-1 to +1)

→ use ~~of~~ for the non-linority



Types of Analysis

1) Univariable → any one feature
2) Bivariable → two feature
3) Multivariable → more than two

1) Univariable Analysis → categorical, → Numerical

1) Categorical feature :- Value counts

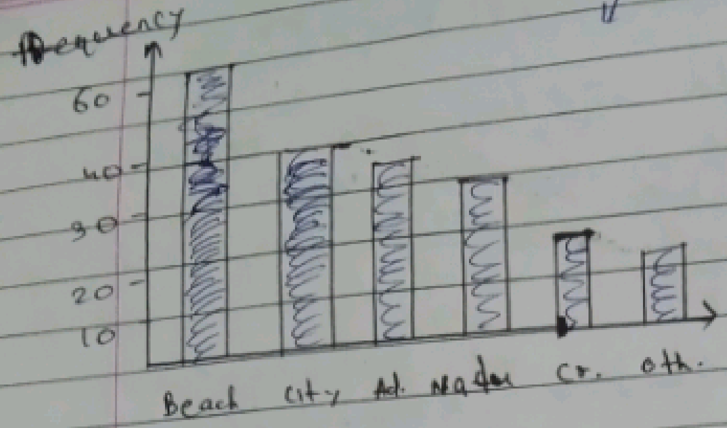
Frequency distribution Table
(frequency ↓ count)

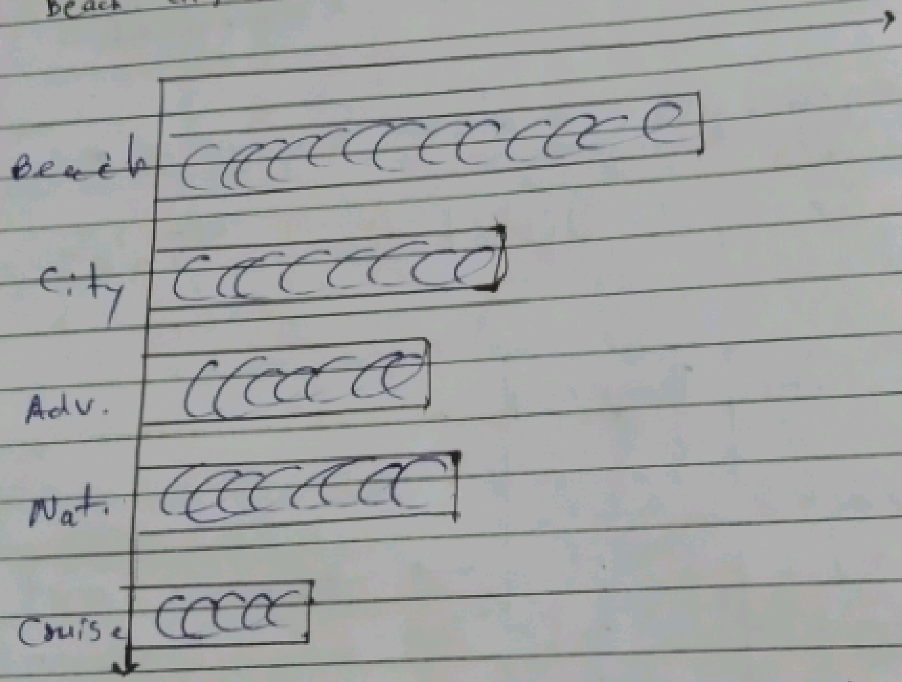
00 t ...dent vacation data

Type of vaccation	Frequency
Beach	60
City	40
Adventure	30
Nature	35
Cruise	20
other	15

→ Bar chart / Column chart

Column chart ✓



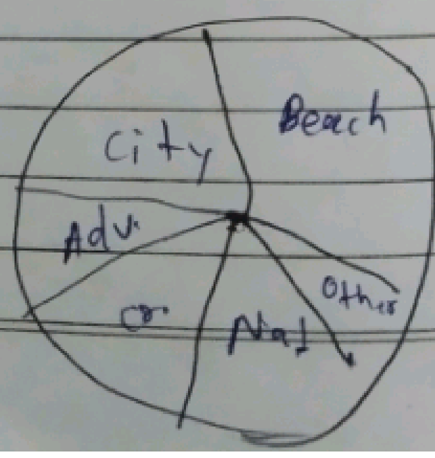


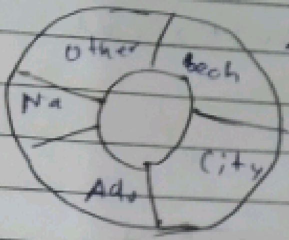
— x —

Distribution Rate → Percentage / Relative frequency

	freq	Relative freq,
Beach	60	60/200 = 0.3 = 30%
City	40	0.2 = 20%
Adv.	30	0.15 = 15%
Nat.	35	0.175 = 17.5%
Other	15	0.075 = 7.5%
City	20	0.1 = 10%

Pie chart





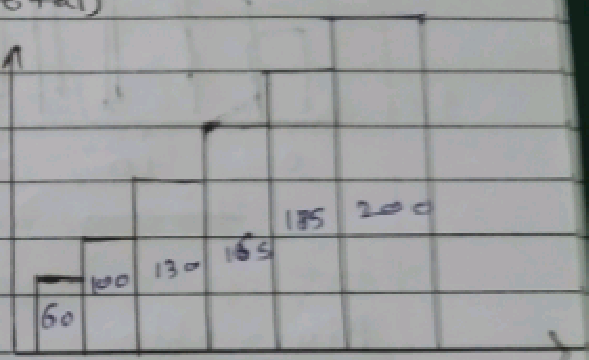
← Donut chart

— x —

Commulative frequency (waterflow chart)

→ (Addition running total)

Beach	60	60
City	40	100
Adve.	30	130
Nat.	35	165
Cr.	20	185
othe.	15	200



— x —

2) **Univariable Analysis** → Numerical Column

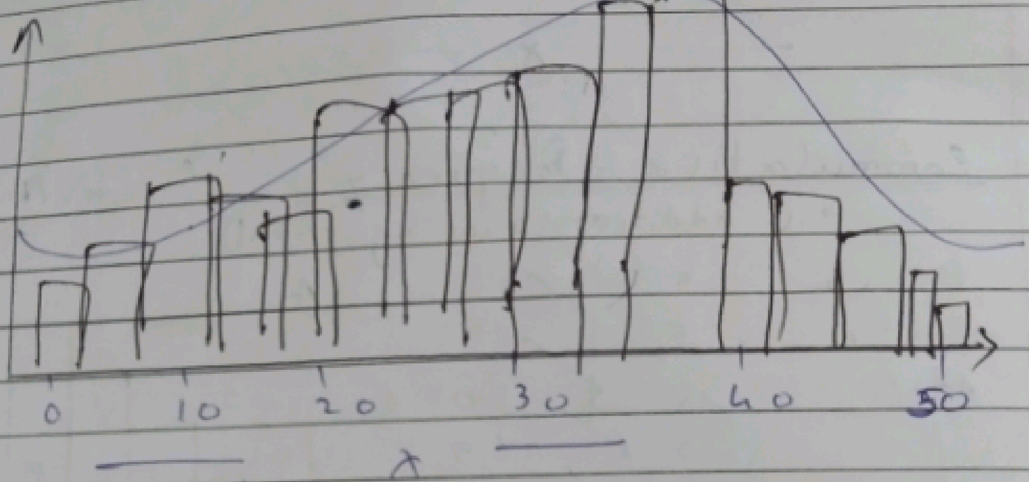
200 Student → categorical conversion

Age	Age group		
8	8 - 18	Teen	10
9	18 - 25	Adult	12
12	25 - 45	Sr. Adult	15
15	45 - 60	old Age	17
28			
25			
9			
8			

Page No:
Date.

Distribution (Numerical Data)

Histogram



Bivariable Analysis → Num vs Cat/ Cat vas Num

↓ Catergorical Vas Cat

↓ Num vs Num

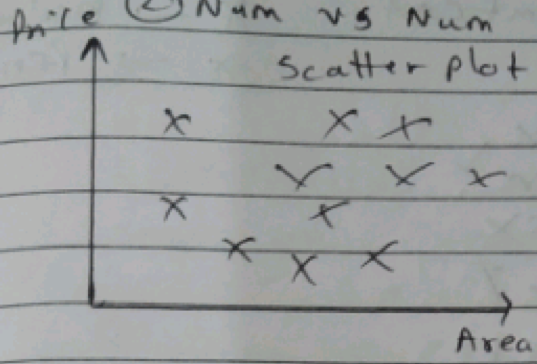
① Cat vs Cat ⇒ Contengency Table/ Cross Tab

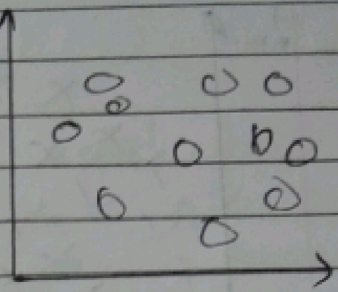
Gender	exam
M	Pass
F	Fail

} 200 student data

Gender	Pass	Fail
M	29	45
F	30	73

② Num vs Num





③ Cat vs Num / Num vs Cat

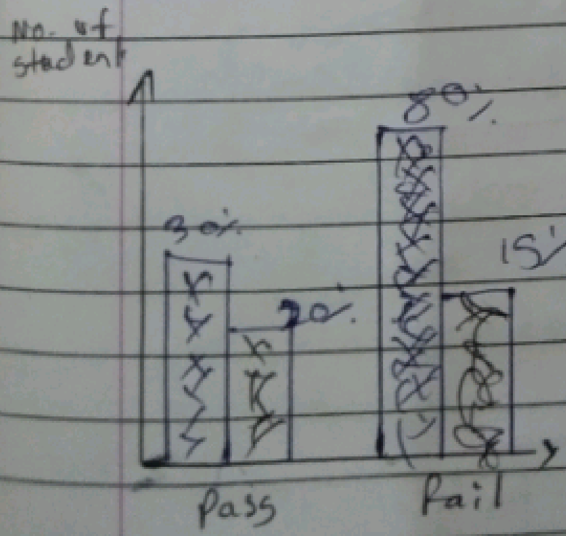
~~Age~~

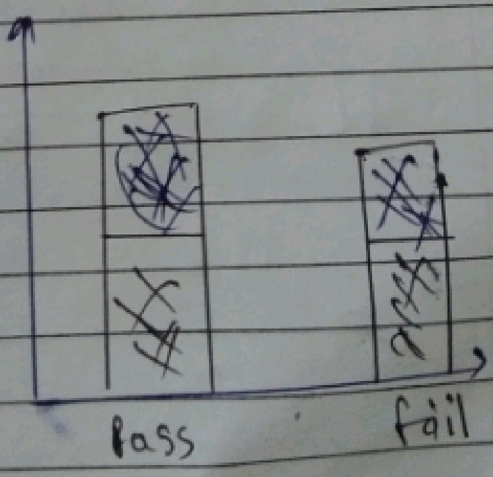
		Age		
male female	Teen	Adult	Sr.	old

~~Categorical~~
Contengency table

Multivariable Analysis

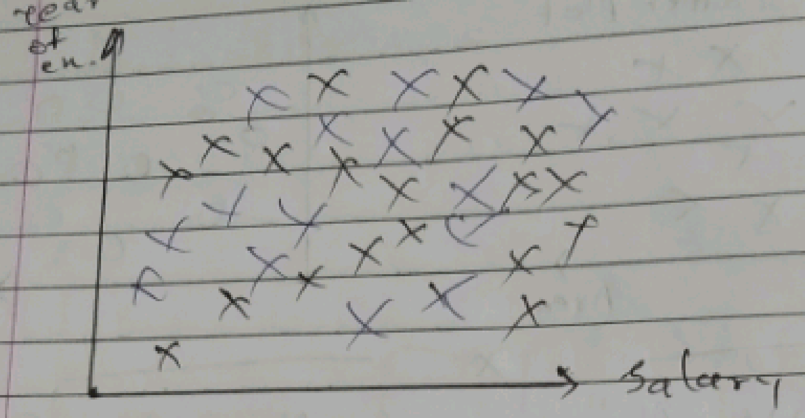
legend ☒ female
☒ male

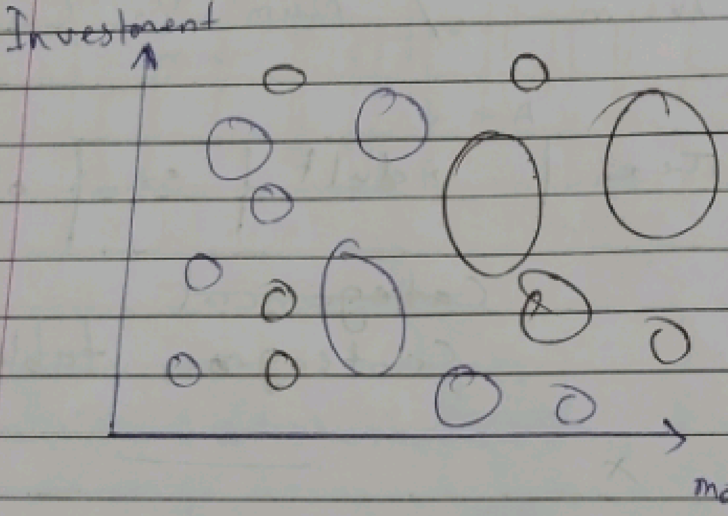




num + num + cat

X → male
x → female





bubble size = Income

O India

O → Remote ↓ outside country